

The pizzetti proofs, explained

Part 3: Crossing the stellar edge (Section 5.3 and Appendix C)

Companion to the pizzetti paper (Sections 5.1, 5.3 and Appendix C)

Abstract

When the planet overlaps the edge of the star, the interior series of Part 1 no longer works. The paper then writes the blocked light as a sum of special functions, turns it into a few short polynomials (“pieces”), and proves an error bound for each piece (Theorem 3, proof in Appendix C). This part explains that proof. Most steps need only first-year calculus. A few need facts from complex analysis, collected in Section 2 as five “tools”, each with the reason it is true and a numerical illustration. With the tools in hand the proof is mainly careful bookkeeping: every quantity is bounded by something computable, and the pieces are added up. Every step is checked in `validation/04_hypergeometric_section5.ipynb` and `validation/05_contact_theorem3.ipynb`.

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1 The plan in plain words

During ingress and egress the planet covers part of the stellar edge. The blocked light $P_\gamma(z, p) = \int (1 - r^2)^\gamma dA$ (over the covered part of the star) still depends only on z and p , but now it has a “kink” at the moment the planet’s edge touches the star’s edge from inside (second contact), and the interior series cannot reach past it. The paper proceeds in three steps:

1. **A formula** (Section 5.1): P_γ is written as an infinite sum over j of known special functions (Gauss hypergeometric functions) of one variable. Keeping J terms gives $P_{\gamma, J}$. *Error 1*: the terms after J .

2. **Pieces** (Section 5.3): the range of z from first contact down to the cut z_c of the interior series is split into five to eight pieces. On each piece $P_{\gamma,J}$ is replaced by a polynomial of low degree N (with an explicit $y^{\gamma+3/2}$ and logarithm factor at second contact). *Error 2*: the polynomial terms after N .
3. **Bounds** (Appendix C): error 1 is bounded by a geometric series (Lemma 2); error 2 by Cauchy's estimate (Lemma 6), which needs an upper bound M for the size of the function on a circle in the complex plane (Lemmas 3–5). The degree N of each piece is the smallest for which errors 1 and 2 together stay below the tolerance τ .

Worked example. Throughout we use the power-2 law $I = 1 - 0.6(1 - \mu^{0.6})$, i.e. exponents $\gamma = 0$ and 0.3 , with $p = 0.1$ and $\tau = 10^{-9}$. The interior series reaches $z_c = 0.8203$. The limb region $0.8203 < z < 1.1$ is covered by six pieces:

piece	range of z	degree N	proven bound
A, first contact	1.0462–1.1000	6	4.4×10^{-10}
MO, middle (in k)	0.9950–1.0462	8	5.2×10^{-10}
MO, middle (in k)	0.9462–0.9950	8	4.7×10^{-10}
B, second contact, outside	0.9000–0.9462	7	5.9×10^{-10}
C, second contact, inside	0.8422–0.9000	12	8.2×10^{-10}
MI, middle (in κ)	0.8203–0.8422	4	2.6×10^{-10}

(Second contact is at $z = 1 - p = 0.9$, first contact at $z = 1 + p = 1.1$.)

2 Five tools from complex analysis

The polynomials are truncated power series. To bound what a truncated power series leaves out, it pays to let the variable be a complex number u . Nothing physical happens there; the complex plane is just where the cleanest bounds live. All five tools are standard (any textbook on complex analysis, e.g. the first chapters of Ahlfors or of Stein & Shakarchi).

Tool 1 (A power series converges in a disk). *A power series $f(u) = \sum_n f_n u^n$ converges inside the largest disk around 0 that contains no “bad point” of f (a point where f is infinite, or where a square root or logarithm has its branch point).*

Example: $1/(1-u) = 1 + u + u^2 + \dots$ converges for $|u| < 1$; the bad point is $u = 1$. The same holds for $\sqrt{1-u}$ or $\ln(1-u)$. This is why each piece is kept small: its expansion point must be far from the bad points of the function, and then the coefficients fall off fast.

Tool 2 (Cauchy's estimate). *If f is given by its power series on and inside the circle $|u| = \rho$, and $|f(u)| \leq M$ on that circle, then*

$$|f_n| \leq M \rho^{-n} \quad \text{for every } n.$$

Why. Put $u = \rho e^{i\theta}$ into the series: $f(\rho e^{i\theta}) = \sum_n f_n \rho^n e^{in\theta}$. This is a Fourier series in θ , so its coefficients are averages over the circle,

$$f_n \rho^n = \frac{1}{2\pi} \int_0^{2\pi} f(\rho e^{i\theta}) e^{-in\theta} d\theta,$$

and an average is never larger than the largest value: $|f_n| \rho^n \leq M$.

What it buys. If we know f is at most M on a circle of radius ρ , then every coefficient is bounded, and the terms $f_n u^n$ with $|u| \leq w < \rho$ are at most $M(w/\rho)^n$: a geometric series whose tail we can sum. Example: $f = 1/(1-u)$ has $|f| \leq 10$ on $|u| = 0.9$, so $|f_n| \leq 10 \cdot 0.9^{-n}$ (true value: $f_n = 1$). Crude, but the factor $(w/\rho)^n$ makes it harmless after a few dozen terms.

Tool 3 (Square roots and powers of complex numbers). *A non-zero complex number has two square roots. On a disk where a function g has no zeros, one can choose $\sqrt{g}$ (and g^a for any real a) so that it varies smoothly, matches the usual positive root on the real axis, and satisfies $|g^a| = |g|^a$. The principal root (the one with positive real part) is such a choice wherever $\text{Re } g > 0$.*

The equality $|g^a| = |g|^a$ is what makes bounds easy: to bound $x^{\gamma+1}$ on a circle, bound $|x|$.

Tool 4 (Largest and smallest values lie on the boundary). *For a function given by a power series, the largest value of $|f|$ over a closed disk is reached on its boundary circle. The real part of such a function takes its smallest and largest values on the boundary as well.*

Why (for the real part). The real part h of such a function has zero Laplacian. By the mean-value expansion of Part 2 with $\Delta h = 0$, its average over any small circle equals its value at the centre. So h cannot have a strict minimum inside the disk: the average around a minimum would be larger than the minimum itself.

Tool 5 (Bounding a hypergeometric series). *The Gauss hypergeometric function is the power series ${}_2F_1(a, b; c; u) = \sum_n t_n u^n$ with $t_0 = 1$ and term ratio*

$$\frac{t_{n+1}}{t_n} = \frac{(n+a)(n+b)}{(n+c)(n+1)}.$$

(a) $|{}_2F_1(u)| \leq \sum_n |t_n| |u|^n$; if all $t_n > 0$, this is ${}_2F_1(|u|)$.

(b) *The term ratio tends to 1, and from some n on it is at most an explicit number $\bar{r}(n)$ that decreases with n (Lemma 5a of the paper). Once $\bar{r}(n)|u| < 1$, the rest of the sum is smaller than a geometric series. So $\sum |t_n||u|^n$ can be computed with a guaranteed upper bound: add terms one by one, then add the geometric tail.*

For (b), the algebra is one line: $(n+a)(n+b) - (n+c)(n+1) = n(a+b-c-1) + ab - c$, so the ratio is $1 + \frac{n(a+b-c-1)+ab-c}{(n+c)(n+1)}$, and each part of the fraction can only shrink as n grows.

One thing we take from the handbook. Near argument 1, a hypergeometric function splits into a smooth part and a part with a non-integer power (and, for half-integer γ , a logarithm):

$${}_2F_1(\dots; 1-y) = A(y) + y^m [B_l(y) \ln y + B_c(y)],$$

where A , B_l , B_c are again power series with term-ratio rules. These “connection formulae” are in every handbook (DLMF §15.8); the coefficients are tabulated by pizzetti and checked against arbitrary-precision evaluations in its test suite. This is the only place where the proof uses a special-function identity without deriving it.

3 The formula: blocked light as a sum of Gauss functions (Section 5.1)

Starting point. By Green’s theorem (the 2D divergence theorem, as in Part 2) the blocked light can be written as an integral along the part of the planet’s rim that lies on the star. With $s = r^2$ running from $a = (z-p)^2$ to s_1 (where $s_1 = 1$ while the planet crosses the edge and $s_1 = (z+p)^2$ while it is inside),

$$P_\gamma = -\frac{1}{2(\gamma+1)} \int_a^{s_1} (1-s)^{\gamma+1} \left[1 + \frac{p^2 - z^2}{s} \right] \frac{ds}{\sqrt{(s-a)(b-s)}}, \quad b = (z+p)^2$$

(paper, Eq. 24). Notebook 4 checks this against the defining area integral to better than 10^{-23} of the stellar flux.

Change of variable. Write $x = 1 - (z - p)^2$ and $s = 1 - x(1 - t)$. Then $s - a = xt$, $b - s = 4zp(1 - kt)$ with

$$k = \frac{x}{4zp} = \frac{1 - (z - p)^2}{4zp},$$

and $ds = x dt$. The variable k runs from 0 at first contact to 1 at second contact; inside the disk we use $\kappa = 1/k$ instead, which runs from 1 at second contact downwards.

The troublesome $1/s$, expanded. Since $1 - s = x(1 - t)$ and $0 \leq x(1 - t) \leq x < 1$ (for $z > p$),

$$\frac{1}{s} = \frac{1}{1 - x(1 - t)} = \sum_{j \geq 0} x^j (1 - t)^j,$$

a geometric series. Integrating term by term, each term is an Euler integral, i.e. a Gauss function of the single variable k (or κ):

$$H_j(k) = \int_0^1 t^{-1/2} (1-t)^{\gamma+1+j} (1-kt)^{-1/2} dt, \quad G_j(\kappa) = \frac{1}{\pi} \int_0^1 t^{-1/2} (1-t)^{-1/2} (1-\kappa t)^{\gamma+1+j} dt.$$

The result (paper, Eqs. 26–27) is

$$P_\gamma = -\frac{x^{\gamma+1}\sqrt{k}}{2(\gamma+1)} \sum_{j \geq 0} c_j H_j(k) \quad (\text{across the edge}), \quad P_\gamma = -\frac{\pi x^{\gamma+1}}{2(\gamma+1)} \sum_{j \geq 0} c_j G_j(\kappa) \quad (\text{inside}),$$

with $c_0 = 1 + p^2 - z^2$ and $c_j = (p^2 - z^2)x^j$ for $j \geq 1$.

Lemma 2 (Error 1): the terms after $j = J$. Look at the integrals defining H_j and G_j . The integrands are positive, and raising j by one multiplies them by $(1-t)$ or $(1-\kappa t)$, numbers between 0 and 1. So the integrals *decrease* with j : $H_j \leq H_{J+1}$ and $\pi G_j \leq \pi G_{J+1} \leq \pi$ for $j > J$. Since $|c_j| = |z^2 - p^2|x^j$, the omitted terms are at most a geometric series in x :

$$|P_\gamma - P_{\gamma,J}| \leq |\text{prefactor}| \cdot |z^2 - p^2| \cdot H_{J+1} \cdot \frac{x^{J+1}}{1-x} \quad (\text{and similarly inside, with } \pi \text{ for } H_{J+1}).$$

On a piece, each factor is largest at one end of the piece (they all move monotonically with z), which gives a single number for the whole piece (paper, Eq. C10).

Worked example. For the first-contact piece A of our example, $J = 7$ terms suffice to make error 1 at most 2.5×10^{-11} of the stellar flux for the exponent $\gamma = 0.3$.

Where this breaks down. The geometric series converges like x^J , and $x = 1 - (z - p)^2$ approaches 1 when the planet's centre comes close to the stellar centre ($z \rightarrow p$). For large planets the pieces nearest to z_c then need too many terms. That is why proven pieces exist only for $p \lesssim 0.2$ (at $\tau = 10^{-9}$); beyond, pizzetti falls back to numerical quadrature and says so.

Check it yourself. `validation/04_hypergeometric_section5.ipynb`: the rim integral and the j -series against the defining area integral (agreement better than 10^{-21}), and Lemma 2 at single points: error/bound between 0.25 and 0.9999, never above 1.

4 The pieces (Sections 5.2–5.3)

Why pieces. By Tool 1 a power series converges up to the nearest bad point. For the blocked light the bad points are the two contacts (where the flux is not smooth) and, inside the disk, $\kappa = 0$. A single series over the whole limb would converge slowly or not at all. Short series around well-chosen centres converge fast.

The five kinds of piece.

- **A (first contact, $k \leq \frac{1}{4}$):** $P_{\gamma,J} = k^{\gamma+3/2} \times (\text{power series in } k)$. The factor $k^{\gamma+3/2}$ shows how the blocked light starts from zero.
- **MO (two middle pieces, $\frac{1}{4} \leq k \leq \frac{3}{4}$):** Taylor series about $k_0 = \frac{3}{8}$ and $\frac{5}{8}$.
- **B and C (second contact, $y = 1 - k \leq \frac{1}{4}$ outside, $y' = 1 - \kappa \leq \frac{1}{4}$ inside):** $P_{\gamma,J} = R(y) + y^{\gamma+3/2} [S_L(y) \ln y + S_C(y)]$, three power series. The non-smooth part at second contact is written out explicitly; the three series are smooth.
- **MI (middle pieces inside, in κ):** Taylor series of half-width at most 0.07, from the C piece down to z_c .

Each piece needs the coefficients of its power series. They follow by multiplying and composing known series (for z as a function of k , for x , for the Gauss functions); this is exact up to the order computed (64). On the middle pieces the series of the Gauss functions about 0 are truncated after a few hundred terms before being re-centred; Section 7 accounts for that.

5 Bounding error 2: Lemma 6 (Cauchy's estimate with a correction)

On a piece with variable u , $|u| \leq w$, write the smooth part as $f(u) = \sum_n f_n u^n$ and use the polynomial $\sum_{n \leq N} f_n u^n$. What is left out is $\sum_{n > N} f_n u^n$. We split it:

- terms $N < n \leq 64$: their coefficients are computed, so add up $|f_n|w^n$ directly;
- terms $n > 64$: Cauchy's estimate (Tool 2) on a circle of radius $\rho > w$ gives $|f_n|w^n \leq M(w/\rho)^n$, a geometric tail $M(w/\rho)^{65}/(1 - w/\rho)$.

This is the paper's Eq. (32)/(C15):

$$\left| \sum_{n > N} f_n u^n \right| \leq \sum_{n=N+1}^{64} |f_n|w^n + M \frac{(w/\rho)^{65}}{1 - w/\rho}.$$

At the contacts, the factors $y^{\gamma+3/2}$ and $y^{\gamma+3/2} |\ln y|$ are replaced by their largest values on the piece (an elementary calculus exercise, Lemma 5d).

Worked example. Piece A, $\gamma = 0.3$: the circle radius chosen is $\rho = 0.87$ against $w = 0.25$, and the bound on the circle is $M \approx 0.2$ (in units of the stellar flux). The far tail is then $0.2 \times (0.25/0.87)^{65}/(1 - 0.29) \approx 10^{-36}$: utterly negligible. The bound of the piece is dominated by the computed terms $N < n \leq 64$ and by error 1; in total 4.4×10^{-10} , while the true error of the piece is 2.4×10^{-10} .

So the only hard part is a *guaranteed* value of M , the size of f on the circle. That is what Lemmas 3–5 provide.

6 Getting M : bounding the function on a circle (Lemmas 3–5)

f is a product of simple factors: the prefactor $(x^{\gamma+1}, \sqrt{k})$, the c_j $(1 + p^2 - z^2$ and $(p^2 - z^2)x^j)$, and the Gauss functions. We bound each on the circle and multiply.

Lemma 3: z and x behave well on the disk. As a function of k the separation is $z = p(1 - 2k) + \sqrt{W}$ with $W = 1 - 4p^2k(1 - k)$ (paper, Eq. 28). Two things could go wrong on a complex disk: the square root could jump between its two values, and z (hence x) could be zero, which would spoil the powers $x^{\gamma+1}$. Neither happens:

- If $4p^2|k||1 - k| < 1$ on the disk, then W lies within distance 1 of the number 1, so $\text{Re } W > 0$, and the principal root is a smooth choice (Tool 3).
- z is never zero, because $(\sqrt{W} + p(1 - 2k))(\sqrt{W} - p(1 - 2k)) = W - p^2(1 - 2k)^2 = 1 - p^2 \neq 0$.

Inside the disk, with κ , the same holds with $V = (\kappa - 2p^2)^2 + 4p^2(1 - p^2)$ in place of W . There the condition $\text{Re } V > 0$ is checked with Tool 4: $\text{Re } V$ is the real part of a power series, so its minimum over the disk lies on the circle, where it is a quadratic in $\cos \theta$ whose minimum is elementary (paper, Eq. C11). The code checks these conditions for every circle it uses and rejects the circle if they fail.

Lemma 4: the size of z and x on the circle, from 1024 samples. Evaluate z at 1024 equally spaced points of the circle and take the largest $|z|$. Between two samples z can be a little larger. How much? By Cauchy's estimate on a slightly larger circle (radius ρ_2), the slope $|z'|$ on the circle is at most $C/(\rho_2 - \rho)$, where C is a crude upper bound of $|z|$ on the larger disk (from the triangle inequality, e.g. $|z| \leq p(1 + 2|k|) + \sqrt{1 + 4p^2|k||1 - k|}$). Every point of the circle is within arc length $\pi\rho/1024$ of a sample, so

$$\max_{\text{circle}} |z| \leq \max_{\text{samples}} |z| + \frac{\pi\rho}{1024} \cdot \frac{C}{\rho_2 - \rho}.$$

Example (piece A): sampling gives $|z| \leq 1.3233$ and $|x| \leq 0.4631$ on the circle; the crude bounds on the larger disk are 1.363 and 0.594; the correction term is tiny.

Lemma 5: the Gauss functions. Three cases, all from Tool 5:

- $H_j(k)$ has positive coefficients, so $|H_j(k)| \leq H_j(|k|)$, a real number computed with a guaranteed upper bound.
- $|G_j(\kappa)| \leq 1$ when $|1 - \kappa| \leq 1$: in its integral, $1 - \kappa t$ lies on the straight line between 1 and $1 - \kappa$, both of size at most 1, and the weight averages to 1.
- At second contact, each of A, B_l, B_c is bounded by the sum of the sizes of its terms, with a geometric tail. For half-integer γ the B_c coefficients contain the digamma function ψ ; the elementary inequality $\ln x - 1/x \leq \psi(x) \leq \ln x$ (for $x \geq \frac{1}{6}$, from Binet's formula) bounds them by a decreasing function of n (Lemma 5b and the paragraph after it).

Multiplying the bounds of all factors and summing over $j \leq J$ gives M (paper, Eq. C13 and the proof of Theorem 3).

7 The middle pieces: one more correction

On the middle pieces the coefficients are obtained by taking the series of each Gauss function about $k = 0$ (or $y = 0$), cutting it after a few hundred terms, and re-centring it at the middle of the piece. Re-centring a polynomial is exact, so what we compute are the exact coefficients of a slightly different function $\tilde{f}$, with the Gauss functions replaced by their truncations. The difference $e = f - \tilde{f}$ is the truncated tail, which Tool 5 bounds on the circle by a number η . Lemma 6 then adds $\eta/(1 - w/\rho)$ to the error (paper, Eq. C16). In practice this term is far below the tolerance; the point is that it is accounted for rather than assumed small.

8 Putting it together: Theorem 3

Theorem 3 (as in the paper, in words). *For $p < \frac{1}{2}$, on every piece whose disk conditions (Lemma 3) hold, the difference between the blocked light P_γ and the piece’s polynomial approximation is at most error 1 (Lemma 2) plus error 2 (Lemma 6 with M from Lemmas 3–5). The flux error is the sum of these over the exponents of the limb-darkening law, weighted by $|c_i|$ and divided by the stellar flux.*

The chain of reasoning, in one table.

Step	Why it holds	Checked in
Blocked light as a sum over j	Green’s theorem; geometric series; Euler integral	notebook 4
Error 1 (terms after J)	integrands shrink with j ; geometric series	notebook 4 (Lemma 2)
Polynomial of the piece	exact series arithmetic to order 64	notebook 5*
Error 2, orders $N + 1$ to 64	coefficients computed	notebook 5*
Error 2, orders above 64	Cauchy’s estimate (Tool 2)	notebook 5*
M : geometry factors	Lemma 3 (Tools 3, 4), Lemma 4 (sampling)	notebook 5*
M : Gauss functions	Lemma 5 (Tool 5, Euler integral, digamma)	notebook 5*
Middle pieces	truncation term η	notebook 5*

* Through the final bound of each piece, not lemma by lemma.

Check it yourself. `validation/05_contact_theorem3.ipynb` evaluates every certified piece on its own, at points spread over its range including both ends, and compares with the defining area integral. In the full run (183 pieces, five laws, p from 0.0092 to 0.2, $\tau = 10^{-9}$ and 10^{-12}) no error exceeded its bound; the largest ratio was 0.98. The bounds are therefore close to the true errors, which also shows they are not wildly pessimistic.

9 What the proof does not cover

- **Rounding.** The bounds are theorems about exact arithmetic. The code evaluates them in double precision and multiplies them by $1 + 10^{-6}$; rounding in the coefficients and in the evaluation of the polynomials is at the level of 10^{-16} of the stellar flux. A bound that is certified down to the last bit would need interval arithmetic.
- **Large planets.** For $p \gtrsim 0.2$ (at $\tau = 10^{-9}$) some pieces cannot be certified (error 1 converges too slowly near $z = p$); those samples are computed by quadrature, and pizzetti reports which samples are covered by the proof.
- **The connection formulae** of the Gauss functions are taken from the handbook (DLMF §15.8) and tested numerically, not re-derived.

10 Summary

Crossing the stellar edge, the blocked light is a sum of Gauss functions of one variable. Cutting that sum after J terms costs at most a geometric series, because the terms shrink. Each of a few pieces then turns the sum into a short polynomial. What that polynomial leaves out is

bounded by computing the first 64 coefficients and bounding the rest with Cauchy's estimate: the coefficients of a function are averages over a circle, so they cannot exceed the function's largest value there. Finding that largest value is the technical core: the geometry is smooth on the circle (Lemma 3), its size is pinned down by sampling plus a slope bound (Lemma 4), and the Gauss functions are bounded by their series or integrals (Lemma 5). None of these steps is deep; there are just many of them, and each is checked numerically.